

14. $u = 2 \sin x + \frac{\cos^2 x}{2 \sin x + \frac{\cos^2 x}{2 \sin x + \frac{\cos^2 x}{2 \sin x + \dots}}}$, where $0 < \cos x < \sin x$ and $\cos x = \frac{1}{3}$. Find the

exact value of u . Write your answer as a single reduced fractional expression.

15. Find the number of distinct real values for x so that $\begin{bmatrix} x-2 & x+5 \\ 2x & x-3 \end{bmatrix}$ will not have an inverse.

16. When David and Jim play pool, the probability that David wins exactly two games out of four is equal to the probability that he wins exactly three games out of four. Find the probability that Jim wins any given game. Write your answer as a common fraction reduced to lowest terms. Assume that the outcome of a game cannot affect the probabilities in other games, and that both players have a non-zero chance of winning a game.

17. A parabola is described by $\begin{cases} x = 2+t \\ y = 1-t^2 \end{cases}$. Find the exact distance between the vertex and the y -intercept of this parabola.

18. Determine the exact sum of all values of θ , $0 \leq \theta < 2\pi$ for which $\cos^2 \theta = \frac{3}{5}$.

19. The function $y = f(x)$ is periodic with period 3. If $f(6) = 4$, $f(8) = 2$ and $f(10) = 9$, find the value of $2f(1) + 3f(2) + 4f(3)$.

20. In a triangle with sides a , b , and c , $(a+b+c)(a+b-c) = 3ab$. Find $\sec C$, where angle C is the angle opposite side c .